Soham Sinha

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Education

University of Massachusetts Amherst — Amherst, MA

B.S. in Computer Science & Mathematics Double Major, Business Minor | ***GPA: 4.0*** — *Expected Graduation: May 2028*

- **Coursework:** Data Structures & Algorithms, Systems Programming (C), SQL & Databases, Machine Learning, Artificial Intelligence, Linear Algebra, Probability & Statistics
- **Awards:** HackHarvard '26 Winner, CMU TartanHacks '26 Winner, MIT Bitcoin Hackathon '26 Finalist, HackImpact '23 Winner, Chancellor's Award, Dean's List (Spring '26, Spring & Fall '25, Fall '24)

Technical Skills

Languages: Python, Java, C, SQL, TypeScript, JavaScript
Frameworks & Data: FastAPI, Node.js, Express, React, PostgreSQL, Redis, WebSockets
Cloud & Infrastructure: AWS (ECS, RDS, S3, ElastiCache, SQS, EventBridge), Docker, Linux
Developer Tools: Git, pytest, JUnit, Claude Code, Codex, Cursor

Experience

Co-Founder & Software Engineer — Feb. 2026 – Present

NullifAI — *Amherst, MA*

- Co-founded a SaaS platform automating digital product feedback, cutting a traditional **2–4 week** workflow to **minutes** at **$1.90 per run** versus an estimated **$300** per campaign.
- Engineered a multi-agent evaluation backend with parallel Playwright browser workers, simulated user personas, structured agent debate, and Pydantic validation to generate repeatable product-feedback reports.

Software Engineer Intern — Jun. 2026 – Aug. 2026

LUME Global — *Part Time — Mumbai, India*

- Architected the backend, PostgreSQL data model, and AWS infrastructure for a platform serving **10,000+ members**, integrating **5 vendor systems** through a **32-table schema** and unified identity model.
- Built and deployed a production FastAPI backend and AI inference workflows on AWS ECS Fargate, using asynchronous database operations, Redis caching, and SQS/EventBridge for event-driven processing.

Software Engineering Intern — May 2026 – Aug. 2026

Siemens Energy — *Remote*

- Built a Python document extraction tool using Gemini Vision API to parse complex, unstructured client PDFs and images into validated JSON with schema enforcement, generating Excel reports and dashboards for **30+ enterprise clients**.
- Engineered structured output parsing with automated validation across diverse document formats, achieving **90% extraction accuracy**; packaged as a standalone executable for direct deployment to client teams.

Machine Learning Intern — Jul. 2025 – Aug. 2025

Finarkein Analytics — *Mumbai, India*

- Optimized BERT fine-tuning using CUDA GPUs and a streamlined training pipeline, reducing training time from **2 hours to 15 minutes** (**8× faster**).
- Built preprocessing and validation pipelines for **100K+ records**, improving BERT text classification accuracy from **67% to 90%** through cleaner training data.

Projects

UMass Housing Search Engine | *Java, Jsoup, JUnit* | Live | GitHub — Aug. 2026 – Sep. 2026

- Architected a concurrent Java crawler consolidating **300+ rental listings** from **7 housing sources** into one searchable index using a 5-thread ExecutorService pool, per-source adapters, rate-limited fetching, and normalized address-price deduplication.
- Engineered a TF-IDF ranked search engine from scratch over a 1,200-term inverted index, serving housing queries in **<8 ms** through a raw Java socket HTTP server and JSON API endpoints. Deployed on Railway.

PulseX | *TartanHacks '26 Winner* | *React, TypeScript, FastAPI, PostgreSQL, WebSockets* | GitHub — Jan. 2026

- Built a FastAPI backend for XRP/RLUSD donations, implementing XRPL escrow and batched payouts with PostgreSQL-backed transaction tracking and real-time WebSocket updates to enable traceable, automated fund allocation.

Activities & Leadership

AI4ALL Ignite Fellow (2026 – 2027): Competitive AI career accelerator with mentorship from OpenAI, Google, and Capital One; selected among **200+ projects** to present ML research at the AI4ALL Symposium.
Head Undergraduate Course Assistant – CICS 240 (Aug. 2025 – Present): Lead UCAs supporting **300+ students** in computational statistics and probability.